University of Colombo, Sri Lanka

University of Colombo School of Computing

Bachelor of Science in Computer Science

First Year Examination — Semester I - 2020/2021

SCS1207 — Software Engineering I

Two (2) Hours

Answer All Questions

Number of Pages = 16 Number of Questions = 4

To be completed by the candidate								
Index Number								

Important instructions to candidates

- Students should answer in the medium of **English language only** using the space provided in this question paper..
- Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- Write your index number **CLEARLY** on each and every page of this Question paper.
- This paper consists of **4** questions in **16** pages (including the Cover Page).
- The duration of the paper is **2** Hours.
- Answer **ALL** questions.
- Programmable/Non-Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.
- Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
3	
4	
Total	

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1. (a). i. Briefly describe **two (2)** reasons that make Software projects different from other projects.

[2 marks]

1.

2.

ii. Briefly describe **two(2)** main reasons for software project failure.

[4 marks]

1.

2.

iii. List down **four (4)** features that should be included in a good software.

[4 marks]

1.

2.

3.

4.

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(b). Identify **all** the **suitable Software Process Model(s)** for the software systems described in **Column A** from the following list and write in **Column B**.

[6 marks]

`{Waterfall Model, Iterative Model, Prototype Model, RAD Model, Incremental Model, Spiral Model}`

	Column A	Column B
1.	Projects with unclear needs or projects still in research and development.	
2.	Systems that need to be produced in a short time and have known requirements.	
3.	Large software that can be easily broken down into modules.	
4.	Projects where the requirements are understood completely and unlikely to radically change.	
5.	Projects that have loosely-coupled parts and projects with complete and clear requirements.	
6	Projects where the customer cannot define requirements clearly and when they are ready to be actively involved and can provide feedback.	

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(c). List down the **three (3)** key activities involved in requirement engineering and describe how each activity is carried out.

[9 marks]

1.   **2.**   **3.**

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2. Agile methods are iterative development methods that reduce process overheads and documentation.

(a). State the core reason for issuing an "Agile Manifesto".

[2 marks]

(b). Select the most appropriate word phrase from the below-listed words to complete the following sentences related to the agile manifesto.

[2 marks]

`[Customer collaboration, Individuals and Interactions, Responding to Change, Working Software]`

i. ______________________________ over processes and tools.

ii. ______________________________ over comprehensive documentation.

iii. ______________________________ over contract negotiation.

iv. ______________________________ over following a plan.

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(c). Briefly explain **three (3)** drawbacks of introducing "Agile methods" in critical systems engineering.

[6 marks]

1.
2.
3.

(d). List **Four (4)** Agile Software Development methods other than SCRUM.

[2 marks]

1.
2.
3.
4.

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(e). Scrum is an agile software development framework that helps teams to work together.

i. List down **three(3)** examples for each of the following in SCRUM.

[5 marks]

A. Scrum Roles

1.

2.

3.

B. Scrum Ceremony

1.

2.

3.

C. Scrum Artifact

1.

2.

3.

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ii. SCRUM is an agile framework that is lightweight, simple to understand, and difficult to master. Do you agree? Justify your answer.

[5 marks]

(f). Briefly explain **three (3)** advantages of "Pair Programming".

[3 marks]

1.

2.

3.

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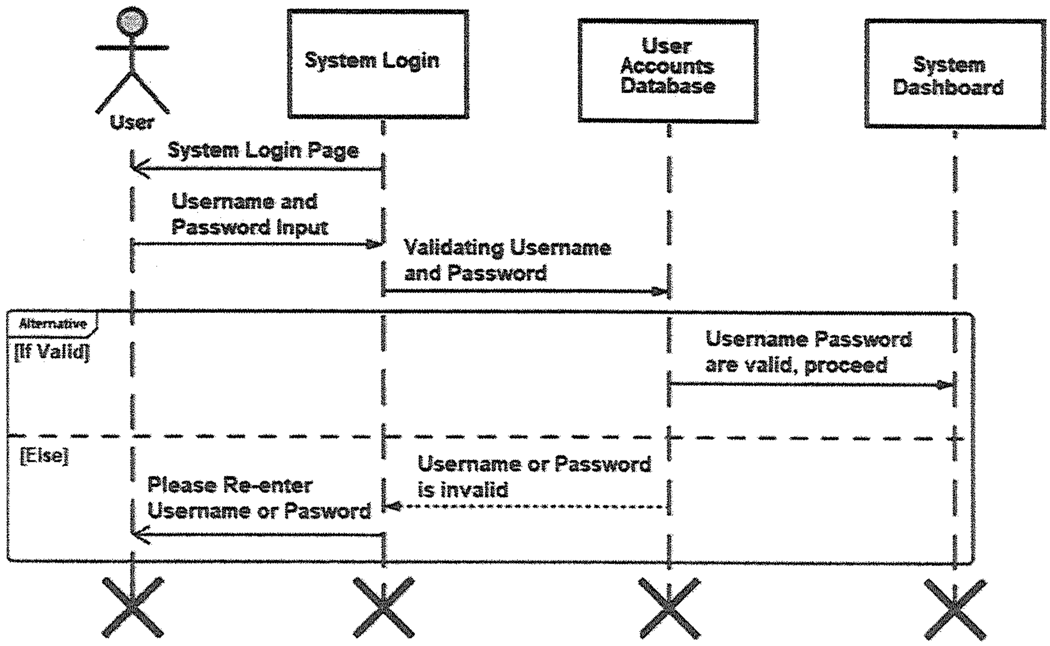
3. (a). Consider the following sequence diagram for a user login.



Answer the follwoing questions based on the above diagram.

i. Identify the actor(s) involved.

[1 marks]

ii. Identify the objects that are in the sequence diagram

[3 marks]

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iii. Explain the given sequence diagram. Your answer should include:
- How the actors involve with the objects
- The flow of the event
- Type of each messege (synchronous, asynchronus, return,reflexive, etc.)
- Alternative scenarios if applicable

[6 marks]

(b). i. Briefly describe what is a software architectural view model using an example.

[3 marks]

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ii. Describe why there should be different architectural views for a particular software.

[4 marks]

(c). i. List down and briefly describe **four (4)** levels of software reuse.

[4 marks]

1.

2.

3.

4.

ii. Explain **two (2)** issues that can arise in software reuse.

[4 marks]

1.

2.

Index Number | | | | | | | | |

4. Late delivery or software defects can damage the company's reputation leading to the loss of customers.

(a). List down the **three (3)** main development testing stages.

[1.5 marks]

1.

2.

3.

(b). Briefly define the below two terms providing each an example.

[4 marks]

Stress Testing

Regression testing

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(c). Compare and Contrast Software Verification and Validation.

[3.5 marks]

	Verification	**Validation**
Similarities		
Differences		

(d). Identify whether the following statements are True or False. In either case the statement should be justified in the space provided.

[5 marks]

i. Static testing techniques can only be applied in software verification processes. (T/F)

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ii. Multiple testing strategies can be applied in testing a single component of a software system that is being developed by the software development teams. (T/F)

iii. Exhaustive defect testing is possible in a program containing loops that can be executed a variable number of times. (T/F)

iv. In test-driven development a regression test suite is developed incrementally as the program is developed. (T/F)

v. White box testing is a common testing technique used in release testing stage. (T/F)

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(e). In an operational software system, change is inevitable if the system has to remain useful. Justify by providing **at least three (3)** facts to support the statement.

[3 marks]

(f). It is always better to reengineer the legacy systems instead of replacing them with a new system.

i. State **three (3)** reasons as to why replacing a Legacy system is risky and expensive.

[3 marks]

1.

2.

3.

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ii. Define the term "Software Reengineering".

[2 marks]

iii. Briefly explain **three (3)** reengineering process activities.

[3 marks]

1.

2.

3.

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